Silicon Institute of Technology

Silicon Hills, Bhubaneswar

| An Autonomous Institute |

4th Semester B.Tech. Mid Term Examination 2022-23

ARTIFICIAL INTELLIGENCE(BTCS-T-PE-999)

BRANCH(S) - Computer Engineering,Computer Science and Technology,Computer Science & Engineering

Duration: 01:30 hr — Full Marks: 25

1 *Answer All*

a Define searching in form of [S, s, O, G]. 1

b Give one example of stochastic and determinstic task environment. 1

c Explain the TURING TEST with the help of a neat block diagram. 1

d State the EVALUATION FUNCTION for uniform cost search and A*. 1

e What is Constraint satisfaction problem (CSP)? Give two real life examples of CSP. 1

f In HILL CLIMBING SEARCH, what are Local Maxima, Global Maxima, Ridges and Plateaux. Use a neat diagram to show each one. 1

2 *Answer All*

a Consider a state space where the start space is number 1 and the successor function for a state n returns two states 2n and 2n+1. Draw a portion of state space for states 1 to 15. 3

Suppose the goal state is 11. List the order in which the nodes will be visited for BFS, DFS, DLS with limit 3 and IDS

b State two advantages and two disadvantages of LSAs over systematic search algorithms discussed before. 3

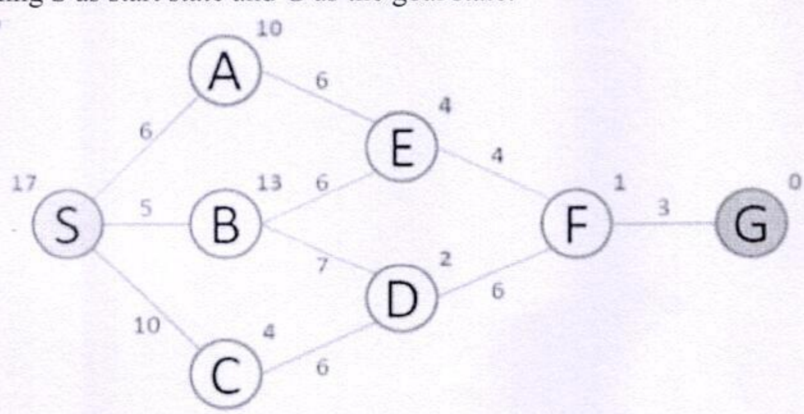
c Explain with a diagram a simple state space where DFS will outperform Iterative depth first search. 3

3 *Answer any One*

a Provide the PEAS descriptions of any three of below intelligent agents: 5

1. Auction Bidding
2. Marathon Running
3. Long Jump
4. Chess Player

b Explain the difference between greedy search and A*. Discuss the steps of traversal of A* search taking S as start state and G as the goal state. 5



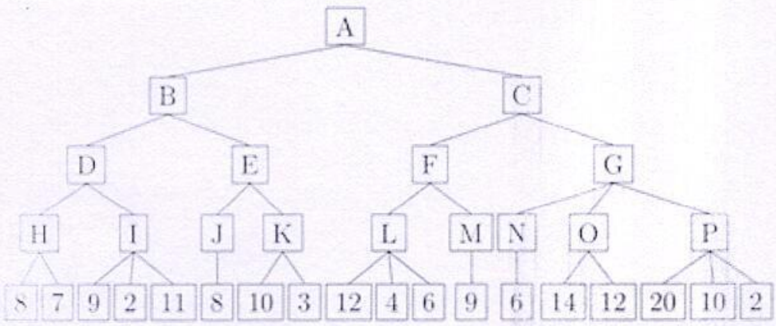
4 ***Answer any One***

a Elaborate the Hill-climbing search algorithm. Find the solution to below 8-puzzle problem using "number of misplaced tiles" heuristics.First diagram is start state and Second diagram is the goal state. 5

1	4	2
	3	5
6	7	8

1		2
3	4	5
6	7	8

b Consider the following tree. Suppose that the top level is a “MAX” node. Use MiniMax algorithm with alpha-beta pruning to determine the values of internal nodes and also indicate which nodes of the tree will be pruned. 5



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4th Semester B.Tech Mid-Term Exam 2023-24

ARTIFICIAL INTELLIGENCE(BTCS-T-PE-999)

BRANCH(S) - Computer Engineering,Computer Science & Engineering,Computer Science and Technology

SIC No.- | | | | | | | | | | |

Duration: 01:30 Total No. of Pages:-02 Full Marks: 25

1 ***Answer All***

a Which uninformed search technique is the more efficient in terms of time and space complexy and why? 2

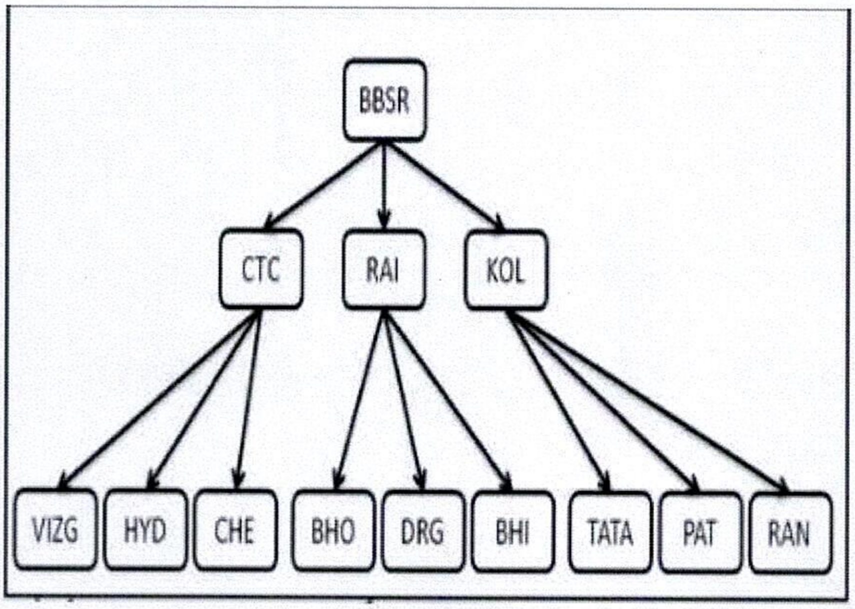
b Illustrate the differences between AGENT FUNCTION and AGENT PROGRAM with automatic vacuum cleaner as example. 2

c Describe the problem definition in Informed Seach in terms of [s, S, O, G, h]. 2

d Describe the HEURISTIC FUNCTION in case of Informed Search. Give examples of any three Heuristic functions. 2

2 ***Answer All***

a Let the Start Location be "BBSR" and Destination be "CHE". Execute the Iterative Deepening DFS Search on below Search Tree. 3.5



b Find the "MANHATTAN DISTANCE" and "NUMBER OF MISPLACED TILES" Heuristics in below 8-Puzzle Example.

Initial State

1	2	3
8		4
7	6	5

Goal State

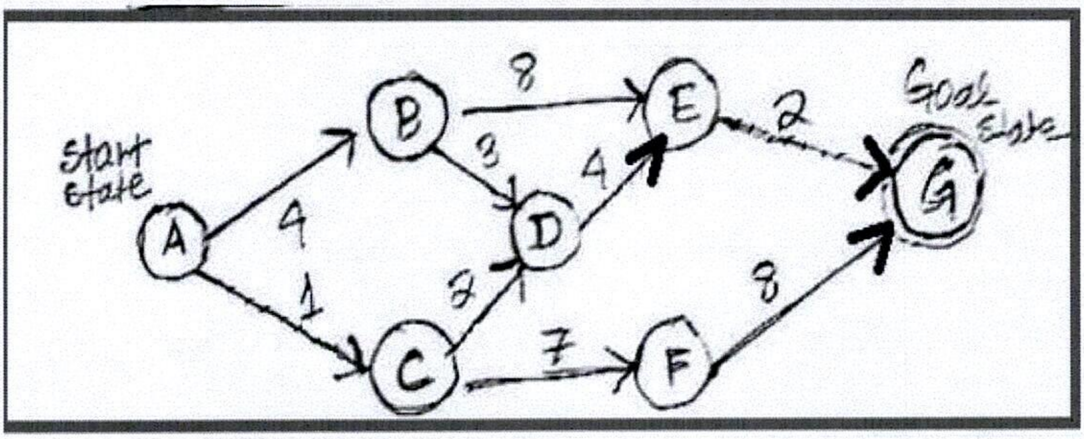
2	8	1
	4	3
7	6	5

[P. T. O.]

3 ***Answer any One***

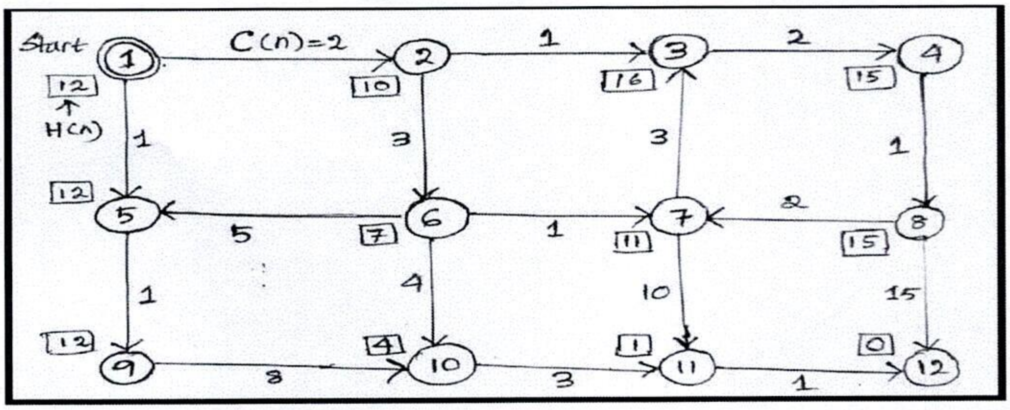
a Describe the "Utility-based Agent" and the "Learning Agent" in detail with block diagrams. Provide examples of both types of agent. 5

b Elucidate the UNIFORM-COST Search approach. What type of data structure is used to simulate the Open or Frontier set. 5
Resolve the below path finding problem from node A to node G.

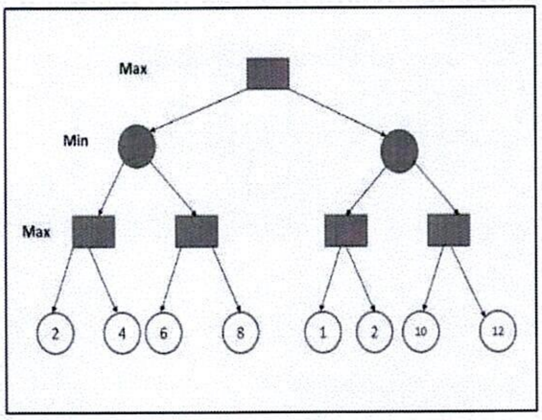


4 ***Answer any One***

a Solve the below path finding problem using A* search algorithm. 5



b Make use of ALPHA-BETA PRUNING apprach to indicate which branches can be pruned in the below problem. 5



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4th Semester B.Tech. Mid Term Examination 2021-22

DESIGN & ANALYSIS OF ALGORITHMS(BTCS-T-PC-011)

Duration: 01:30 | Full Marks: 25

1 ***Answer All***

a Write the recurrence equation for the worst case behaviour QUICK sort. 1

b Show that $\sum_{i=0}^{n} i = \theta(n^2)$ 1

c Order the following functions by decreasing order of asymptotic growth: 1
n^2, $2^{\lg n}$, lg (n!), 100000, n^3, n lg n

d Is an array that is in sorted order a min heap? 1

e In the array representation of an n-element MIN-HEAP what is the index of first leaf node? 1

f Working modulo q=11, how many spurious hits does the Rabin-Karp matcher encounter in the text T=31415926 when looking for the pattern P=26. 1

2 ***Answer any Three***

a Explain the Divide-and-Conquer technique. Design a recursive algorithm for Binary Search. 3

b Solve the following recurrence: 3

$$T(n) = 7T\left(\frac{n}{2}\right) + n^2$$

c Explain the HEAP-INCREASE-KEY operation of the priority queue using heap with a suitable example and find its time complexity? 3

d Illustrate the operation of MAX-HEAPIFY(A, 3) on the array A=<27, 17, 3, 16, 13, 10, 1, 5, 7, 12, 4, 8, 9, 10>. 3

3 ***Answer any One***

a Illustrate the operation of PARTITION on the array A = <13, 19, 9, 5, 12, 8, 7, 4, 11, 2, 6, 21>. 5
Show that the running time of QUICKSORT is $\theta(n^2)$ when the array A contains distinct elements and is sorted in decreasing order.

b Write down the algorithm for MERGE-SORT and MERGE procedure. Show that the running time of merge sort is O (n lg n). 5

4 ***Answer any One***

a Find the optimal parenthesization of matrix-chain product whose sequence of dimension is <5, 10, 3, 12, 5, 50, 6>. 5

b Find the LCS from the given sequences X = {A, B, C, B, D, A, B} and Y = {B, D, C, A, B, A} using tabular method of Dynamic Programming. 5

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Total No. of Pages:-**01**

4th Semester B.Tech. Mid Term Examination 2022-23

DESIGN & ANALYSIS OF ALGORITHMS(BTCS-T-PC-011)

BRANCH(S) - Computer Engineering,Computer Science and Technology,Computer Science & Engineering

Duration: 01:30 hr | Full Marks: 25

1 *Answer All*

a Differentiate between Big-Oh notation and Little-Oh notation. 1

b Justify your answer: Is $2^{2n} = O(2^n)$? 1

c When a function is called polynomially bounded. Is the function **(⌊log(n)⌋)!** polynomially bounded ? 1

d What is the height of an n-element HEAP? 1

e When a product of matrices is called fully parenthesized. In how many different ways a chain of 4 matrices <A1, A2, A3, A4> can be fully parenthesized. 1

f What are the elements of Dynamic Programming? 1

2 *Answer All*

a Solve the following recurrence: 3

$$T(n) = T\left(\frac{9n}{10}\right) + n$$

b Order the following functions by decreasing order of their asymptotic growth: 3
$2^{lg\,n}, lg(n!), 10^6, n!, n^3, n^{lg\,n}, 2^n, lg^2 n$

c Show that any comparison based sorting algorithm requires **Ω(n lg n)** comparisons in worst case to sort **n** elements. 3

3 *Answer any One*

a When Master's method fails to solve a recurrence relation of the form 5
$T(n) = a\,T\left(\frac{n}{b}\right) + f(n)$, with $a \geq 1$ and $b > 1$. Justify your answer with the recurrence relation : $T(n) = 2\,T\left(\frac{n}{2}\right) + \frac{n}{lg\,n}$, also solve it using other suitable method.

b Write the QUICKSORT algorithm along with the PARTITION procedure. Discuss when the QUICKSORT algorithm will attain worst-case running time. Write the recurrence and solve it to find the worst case running time of QUICKSORT. 5

4 *Answer any One*

a Write the **MAX-HEAPIFY(A, i)** algorithm and show that the running time of MAX-HEAPIFY on a subtree of size **n** rooted at a given node i is **O(lg n)** 5

b Define matrix chain multiplication problem. Write the algorithm for computing m and s table. Find the m and s table computed by the algorithm for the following matrix dimensions: A1(25,35), A2(35,15), A3(15,5), A4(5, 40) 5

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4th Semester B.Tech Mid-Term Exam 2023-24

DESIGN & ANALYSIS OF ALGORITHMS(BTCS-T-PC-011)

BRANCH(S) - Computer Engineering,Computer Science and Technology,Computer Science & Engineering

SIC No.-

Duration: 01:30 | Total No. of Page:-01 | Full Marks: 25

1 ***Answer All***

a Define the asymptotic notation theta(Θ) 1

b Order the following functions by decreasing order of asymptotic growth: 1

n^2, $2^{\lg n}$, lg (n!), 100000, n^3, n lg n

c What is the running time of the following code segment: 1

```
int a = 0, i = N;
while (i > 0)
{
   a += i;
   i /= 2;
}
```

d what is asymptotic tight bound of BUILDMAXHEAP() procedure? 1

e What are the differences between Dynamic Programming and Divide and Conquer paradigm? 1

f What are the minimum and maximum number of elements in a heap of height h? 1

2 ***Answer any Three***

a Solve recurrenece relation: 3

$T(n) = 2T(\sqrt{n}) + lg\, n$

b Solve the following recurrence : T(n) = 3 T(n/4) + n^2 3

c Explain the HEAP-INCREASE-KEY operation of the priority queue using heap with a suitable example and find out its time complexity? 3

d Show that any comparison based sorting algorithm requires **Ω(n lg n)** comparisons in worst case to sort **n** elements. 3

3 ***Answer any One***

a Write down the algorithm for MERGE-SORT and MERGE procedure. Show that the running time of merge sort is O (n lg n). 5

b Write down the algorithms for QUICK-SORT and PARTITION procedure. Find the best case and worst case time of QUICK-SORT. 5

4 ***Answer any One***

a Write the MAX-HEAPIFY(A, i) algorithm and show that the running time of MAX-HEAPIFY on a subtree of size n rooted at a given node i is O(lg n) 5

b Determine an LCS of <1, 0, 0, 1, 0, 1, 0, 1> and <0, 1, 0, 1, 1, 0, 1, 1, 0> using the tabular method of Dynamic Programming 5

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4th Semester B.Tech. Mid Term Examination 2021-22

DATABASE MANAGEMENT SYSTEMS(BTCS-T-PC-009)

Duration: 01:30 Full Marks: 25

1 ***Answer All***

a What is a database schema? How does it differ from database instance? 1

b What is identifying relationship? 1

c What do you mean by complex attribute? Give example 1

d Differentiate between disjoint and overlapping constraints in a generalization. 1

e With example differentiate between procedural and non-procedural query language. 1

f With suitable examples, differentiate between SELECT operation in SQL and PROJECT operation in Relation Algebra. 1

2 ***Answer any Three***

a Explain the concept of participation constraints with the help of a suitable ER diagram. 3

b Define Data Model. Mention the various categories of Data Models. 3

c Given: R: 3

A	B	C
x	1	2
y	2	3
z	3	4

S:

D	C	F
x	3	4
z	2	3

Evaluate the following relational algebra operators :
a) R U S, b) R - S, c) R X S, d) R |X| S

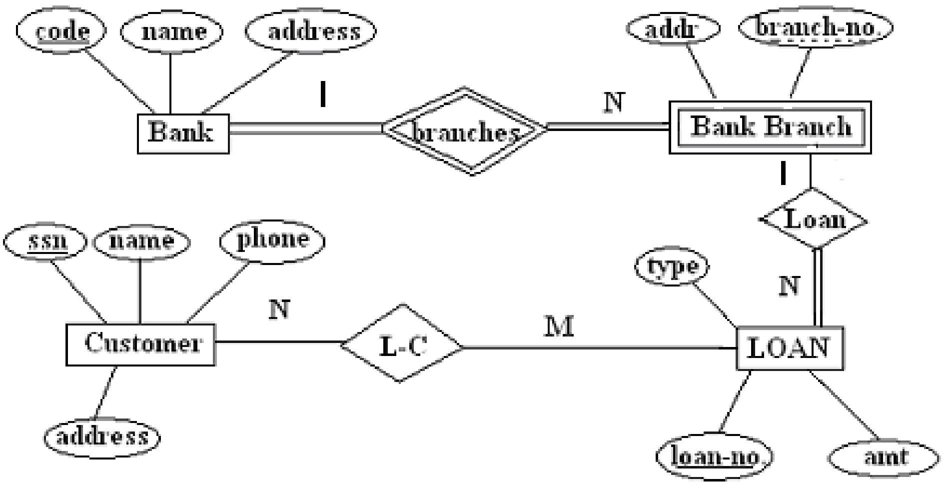
d Consider two relations R1(A,B) with the tuples (1.5), (3,7) and R2(A,C) with tuples (1,7), (4,9).
Compute left outer join, right outer join, and full outer join of R1 and R2. 3

3 ***Answer any One***

a Suppose that you are designing a schema to record information about **reality shows on TV**. Your database needs to record the following information: 5
For each reality show, its name, gender, basic information, and participants' name. Any reality show has at least two or more participants. For each producer, the company name, and company country. A show is produced by exactly one producer. And one producer produces exactly one show. For each television, its name, start year, head office. A television may broadcast multiple shows. Each show is broadcasted by exactly one television. Each user has a username, password, and age. A user may rate multiple shows, and a show may be rated by multiple users. Each rating has a score of 0 to 10.
Draw an entity-relationship diagram for this database.

[PTO]

b Convert the following ER-diagram into relational model. 5



4 ***Answer any One***

a Consider the following relational schema in which the key fields are underlined: 5

SAILOR (**sid**, sname, rating, age)
BOAT (**bid**, bname, colour)
RESERVE (**sid, bid**, day)
Write the following queries in TRC and DRC:
a) Find the name and id of the sailors who have reserved a yellow and a blue colour boat?
b) Find the sailors who have not reserved any boats?
c) Find the IDs of sailors with age over 42 and who have not reserved any green colour boat?
d) Find the colours of boats reserved by Ramesh?

b EMP(name, **ssn**, bdate, sal, superssn, dno) 5
DEPARTMENT(dname, **dnumber**, mgrssn, mgrstartdate)
PROJECT(pname, **pnumber**, dnum)
WORKS_ON(**ssn, pno**, hours)

Write the relational algebraic expression for the following queries:
a) List the project with project number whose department name is 'Research'.
b) List the name of employees who are working in the same department where "Mr. Pradeep" is working.
c) Find the name of the project which is under the supervision of the "Research" department whose mgrssn=12345.
d) Find the name of an employee whose salary is greater than the average salary of each department.

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Total No. of Pages:-**02**

4th Semester B.Tech. Mid Term Examination 2022-23

DATABASE MANAGEMENT SYSTEMS(BTCS-T-PC-009)

BRANCH(S) - Computer Engineering,Computer Science and Technology,Computer Science & Engineering

Duration: 01:30 hr Full Marks: 25

1 *Answer All*

a What is a schema? How is it different from a sub-schema? 1

b Find the cardinality and degree of relation R given below. 1

Roll	name	age
1	smith	30
4	jones	60
7	kumar	90

c Can we apply R union S on the tables given below. If yes find R union S else explain. 1

R

Name	Age
Ram	25
Kumar	36

S

Name	Age	Sex
Ram	25	M
Sita	36	F

d Differentiate between equi join and natural join. 1

e Given, R (A, B, C) is a relation with its attributes. Convert the following relational algebra operations into TRC. 1
$\Pi_{A,B}(R)$

f What is materialized evaluation of query? 1

2 *Answer All*

a With a suitable example, explain what do you mean by referential integrity constraint. 3

b Describe three schema architecture with diagram. 3

c Given: 3

R:

A	B	C
x	1	2
y	2	3
z	3	4

S:

D	C	F
x	3	4
z	2	3

Evaluate the following relational algebra operators :
a) R U S, b) R - S, c) R X S, d) R |X| S

PTO

3 ***Answer any One***

a Consider a STUDENT database system. Think about a possible problem statement, and then construct an ER-diagram to consist of a minimum of four valid *entity sets* for the given database system. Include all the different types of *attributes* and *constraints* to the entity sets and associations that can be possible. Finally, convert the ER model into Relational model. 5

b Draw the system structrure of DBMS. Explain the functionalities of the following components of DBMS: DDL Compiler, Query Compiler, precompiler, runtime database processor. 5

4 ***Answer any One***

a Specify the following queries on the database schema shown below using SQL and Relational algebra expression. 5
Employee(Fname,Mname,Lname,Eno,Dob,Address,Sex,Salary,Supereno,Dno)
Department(Dname,Dumber,Mgreno,Mgrstartdate)
Project(Pname,Pnumber,Plocation,Dnum)
Works_on(Eeno,Pno,Hours)
Depedent(Eeno,Depedentname,sex,dob,Relationship)

(i). Retrieve the average salary of all female employess.
(ii).List the names of all employees who have a depedent with the same first name as themselves.

b Apply heuristic optimization to solve the following query that is based on the following database schema. 5
Employee (SSN, Name, Bdate, Address)
Works_On (ESSN, Pno)
Project (Pnumber, Pname)
Query: Find the names of Employees born after 2001 who work on a project named "Ugly Fox".

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4th Semester B.Tech Mid-Term Exam 2023-24

DATABASE MANAGEMENT SYSTEMS(BTCS-T-PC-009)

BRANCH(S) - Computer Engineering,Computer Science and Technology,Computer Science & Engineering

SIC No.-

Duration: 01:30 | Total No. of Pages:-02 | Full Marks: 25

1 ***Answer All***

a Differentiate between a strong entity set and a weak entity set with examples 1

b What is a data dictionary? Write its importance. 1

c Differentiate between the primary and foreign keys of a relation. 1

d What is a query tree? How does it represent a relational algebraic expression? 1

e What is Equijoin? How is it different from a natural join operation? 1

f Consider the relations $r_1(P, Q, R)$ and $r_2(R, S, T)$ with primary keys P and R respectively. The relation r_1 contains 2000 tuples and r_2 contains 2500 tuples. What is the maximum size/cardinality of the join operation? Explain. 1

2 ***Answer All***

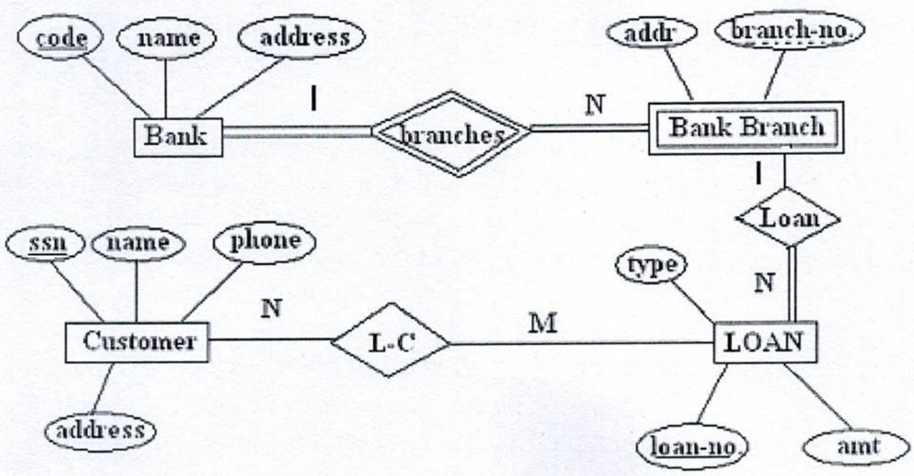
a Describe three schema architectures with a diagram. 3

b Define the following terms: Domain constraint, Entity integrity, and Referential integrity. 3

c Consider two relations R1(A, B) with the tuples (1,5), (3,7) and R2(A, C) with tuples (1,7), (4,9).
Compute left outer join, right outer join, and full outer join of R1 and R2. 3

3 ***Answer any One***

a Convert the following ER diagram into a relational model. 5



b Suppose you are given the following requirements for a simple database for the National Hockey League (NHL): 5

i) The NHL has many teams,
ii) Each team has a unique name, a city, a coach, a captain, and a set of players,
iii) Each player belongs to only one team,
iv) Each player has a unique ID, a name, a position (such as left-wing or goalie), a skill level, and a set of injury records,
v) A team captain is also a player,
vi) A game is played between two teams referred to as host_team and guest_team
vii) The date of the game and the score of the game are recorded.

Construct a clean and concise ER diagram for the NHL database to include all the mentioned requirements.

4 ***Answer any One***

a Consider the following relations and answer any two queries in relational algebra and SQL. 5

Suppliers(SID, Sname, Rating), Parts(PID, Pname,Color), Catalog(SID, PID, Cost)

a) Retrieve SIDs and supplier names of Suppliers whose rating is more than 5.
b) Find names of dealers that supply only red and blue parts.
c) Find the name of the parts whose cost is more than Rs 1000.00.

b By considering the following schema, optimize the query given below. 5

Supplier(Sid,name,address)

Parts(pid,pname,coloar)

catalog(sid,pid,cost)

Find the names of suppliers who supply some red or green parts.

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4th Semester B.Tech. Mid Term Examination 2021-22

ENGINEERING ECONOMICS(BTBS-T-HS-018)

Duration: 01:30 Full Marks: 25

1 ***Answer All***

a What are the three basic problems of an economy? 1

b Why money has time value? 1

c What is sinking fund factor? 1

d What is capitalized Cost? 1

e Define IRR. 1

f What do you mean by depreciable property? 1

2 ***Answer All***

a A person is just 30 years old. He plans to invest an equal sum of Rs. 20,000 every year for the next 30 years from the end of next year. The bank gives 10 % interest compounded annually. Find the maturity value of his account when he is 60 years old. 3

b What do you mean by pay-back period comparison? Give a numerical example to explain the concept. 3

c A company takes a loan of Rs. 20,00,000 to modernise its boiler section. The loan is to be repaid in 20 equal instalments at a 12 % interest rate, compounded annually. Find the equal instalment amount that should be paid for the next 20 years 3

3 ***Answer any One***

a Find the best alternative using the present-worth method of comparison. Assume an interest rate of 15% compounded annually. 5

Alternative	A	B
Initial Cost(Rs.)	6,00,000	8,00,000
Annual receipt(Rs.)	3,00,000	2,00,000
Life (Years)	4	6
Salvage Value (Rs.)	1,00,000	50,000

b Machine A has a first cost of Rs.60,000, no salvage value at the end of its 4-year useful life, and annual operating costs of Rs.20,000. Machine B costs Rs. 90,000 new and has an expected resale value of Rs. 25,000 at the end of its 5-year economic life.Operating costs for machine B are Rs. 24,000 per year. Compare the two alternatives on the basis of their present worths, using the repeated-projects assumptions at 10 percent annual interest. 5

[P. T. O.]

4 ***Answer any One***

a Consider the following cash flow of a project: 5

Year	0	1	2	3	4	5
Cash Flow	-10,000	4,000	4,500	5,000	5,500	6,000

Find the rate of return of the project

b A company invests in one of the two mutually exclusive alternatives. The life of both alternatives is estimated to be 5 years with the following investments, annual returns and salvage values. 5

Particulars	**A**	**B**
Investment (Rs)	1,50,000	1,75,000
Annual equal return (Rs)	60,000	70,000
Salvage Value (Rs.)	15,000	35,000

Determine the best alternative based on the annual equivalent method by assuming i= 25%

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4th Semester B.Tech Mid-Term Exam 2023-24

ENGINEERING ECONOMICS(BTBS-T-HS-018)

BRANCH(S) - Computer Engineering,Computer Science and Technology,Computer Science & Engineering

SIC No.-

Duration: 01:30 Total No. of Pages:-02 Full Marks: 25

1 *Answer All*

a What do you mean by the time value of money? 1

b What is an arithmetic gradient series? 1

c What is capitalized cost? 1

d What is the difference between the capital-recovery factor and the sinking-fund factor? 1

e Distinguish between the ownership life and the economic life of the asset. 1

f Define IRR. 1

2 *Answer All*

a A project has an initial cost of Rs.100, 000 and uniform annual benefits of Rs.12, 500. At the end of its 8-year useful life, its salvage value is Rs. 30,000. What is the net present worth of the project at an interest of 10 percent compounded annually? 3

b A company deposits Rs.20,000 in a bank account at the end of every year for 10 years. If the bank pays 8%, how much is the account worth at the end of 10 years? 3

c A factory purchased a new tooling machine for Rs.20, 500. It is projected that it will have a useful life of 9 years with a Rs.5, 000 salvage value at the end of the 9 years. Assuming a 10% interest rate, find the equivalent uniform annual cost of the machine. 3

3 *Answer any One*

a 5

	Initial Cost (Rs.)	Annual Revenue (Rs.)	Annual Operation and maintenance cost(Rs.)	Salvage Value (Rs.)	Life Year
Project X	4,00,000	1,75,000	50,000	1,00,000	4
Project Y	7,00,000	2,50,000	70,000	1,50,000	6

Select the best project by using the present worth method repeatability assumption with an interest rate of 13 percent.

b Use the PW Method repeatability assumption to select the best project . 5

	Project A	Project B
Initial Cost	2,00,000	3,00,000
Annual Income	30,000	45,000
Salvage Value	50,000	70,000
Useful Life	10 years	15 years
Interest Rate	12%	12%

4 ***Answer any One***

a Find IRR from the following data. 5

Initial Cost: Rs. 2,50,000

Annual Revenues: Rs. 1, 00,000

Annual Expenses: Rs. 30,000

Salvage Value: Rs.50,000

Useful life : 5 years

b 5

Year	0	1	3	4	5
Cash-Flows	-50	-10	30	40	50

Find IRR

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4th Semester B.Tech Mid-Term Exam 2023-24

MATHEMATICS-IV for Computer Sciences(BTBS-T-BS-018)

BRANCH(S) - Computer Engineering,Computer Science and Technology,Computer Science & Engineering

SIC No.- | | | | | | | | | | |

Duration: 01:30 | Total No. of Pages:-02 | Full Marks: 25

1 ***Answer All***

a If the LPP is Maximize Z = CX subject to AX≤b, X≥0 and A be an 4x6 matrix , then how many variables are required for initial simplex table? 1

b When the Simplex table indicates that the LPP has an alternative solution ? 1

c How many basic solutions exist in a LPP with 3 constraints and 5 variables.? 1

d How do you find infeasible solution in dual simplex method ? 1

e Answer the following 1

(a) If the primal variable x_j is unrestricted in sign , then which sign the corresponding dual constraint will hold ?

(b) In which condition, the dual simplex method is applicable in initial table ?

2 ***Answer any Two***

a Farmer Jones must determine how many acres of corn and wheat to plant this year. An acre of wheat yields 25 bushels of wheat and requires 10 hours of labor per week.An acre of corn yields 10 bushels of corn and requires 4 hours of labor per week. All wheat can be sold at $4 a bushel, and all corn can be sold at $3 a bushel. Seven acres of land and 40 hours per week of labor are available.Government regulations require that at least 30 bushels of corn be produced during the current year. Formulate an LPP and find the solution to maximize the total revenue . 2.5

b Solve the following using Simplex method 2.5

$$
\begin{aligned}
&.Maximize\ Z = -x_1 + 4x_2 \\
&subject\ to\quad x_1 - x_2 \leq 1 \\
&\qquad\qquad -2x_1 + x_2 \leq 2 \\
&\qquad and\ x_1,\ x_2 \geq 0
\end{aligned}
$$

c Find the basic feasible solutions from the equations,if exist 2.5

$$
\begin{aligned}
x_1 + 5x_2 + 2x_3 &= 7 \\
2x_1 + 3x_2 + 4x_3 &= 11
\end{aligned}
$$

3 ***Answer any Two***

a 2.5

	Cj	2	3	1	0	0	
CB	Basis	x1	x2	x3	x4	x5	RHS
2	x1	1	0	-1	4	-1	1
3	x2	0	1	2	-1	1	2
	C(bar) row	0	0	-3	-5	-1	Z=8

Maximize Z=Cx ,subject to Ax=b ,x>=0 has the optimal table given above.

Assume (x4,x5) was the initial basis.

How much can C2 be changed ,so that current optimal solution remains optimal ? **[PTO]** b Use revised simplex method find **the initial table** of the given linear programming problem : 2.5

$$Maximize\ z = 2x_1 + x_2$$
$$subject\ to\ \ 3x_1 + 4x_2 \le 6$$
$$6\,x_1 + x_2 \le 3$$
$$x_1, x_2 \ge 0$$

c Write the dual of the following primal problem 2.5

$$Minimize\ Z = 2x_1 + x_2$$
$$subject\ to,$$
$$x_1 - 3x_2 \le 5$$
$$4x_1 + x_2 \ge 7$$
$$and\ \ x_1\ is\ unresticted\ in\ sign\ ,\ x_2 \ge 0$$

4 *Answer any One*

a Solve the given LPP by simplex method. 5

Maximize: $z = 2x_1 - x_2 + x_3;$

Subject to

$3x_1 - 2x_2 + 2x_3 \le 15;\ -x_1 + x_2 + x_3 \le 3;\ x_1 - x_2 + x_3 \le 4;\ x_j \ge 0\ (j = 1,\ 2,\ 3)$

b Solve the following LPP using the Big-M method 5

$$Maximize\ Z = 4x_1 + 5x_2 - 3x_3$$
$$subject\ to,\ \ x_1 + x_2 + x_3 = 10$$
$$x_1 - x_2 \ge 1$$
$$2x_1 + 3x_2 + x_3 \le 40$$
$$and\ \ x_1, x_2, x_3 \ge 0$$

5 *Answer any One*

a Solve the following integer linear programming problem using Branch and Bound method : 5

$$Maximize\ Z = 2x_1 + x_2$$
$$subject\ to\ \ 2x_1 + 5x_2 \le 17$$
$$3x_1 + 2x_2 \le 10$$
$$x_1, x_2 \ge 0\ \ and\ integers$$

b Use dual simplex method to solve the givn LPP 5

Minimize $z = x_1 + 2x_2 + 3x_3$

Subject to $x_1 - x_2 + x_3 \ge 4;\ x_1 + x_2 + 2x_3 \le 8;\ x_1 - x_2 \ge 2,\ x_j \ge 0\ (j = 1,\ 2,\ 3)$

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4th Semester B.Tech. Mid Term Examination 2021-22

MATHEMATICS-IV FOR COMPUTER SCIENCES(BTBS-T-BS-019)

Duration: 01:30 Full Marks: 25

1 ***Answer All***

a Given a linear programming problem with system of m linear equations with n variables $(m < n)$. Define its basic solution, basic variables and non-basic variables. 1

b How to know an unbounded solution exist in a simplex method ? 1

c How many basic solutions exist in a LPP with 3 constraints and 5 variables.? 1

d Answer the following 1

(a) If the primal variable x_j is unrestricted in sign , then which sign the corresponding dual constraint will hold ?

(b) In which condition, the dual simplex method is applicable in initial table ?

e Under what condition does the Dual simplex Algorithm fails ? 1

f If the simplex table is optimal but infeasible, then which method will give us feasible solution? 1

2 ***Answer All***

a Find solution using graphical method 3
MAX $z = 10x_1 + 6x_2$
subject to,
$5x_1 + 3x_2 \leq 30$
$x_1 + 2x_2 \leq 18$
and $x_1, x_2 \geq 0$

b An airline offers coach and first-class tickets. For the airline to be profitable, it must sell a minimum of 25 first-class tickets and a minimum of 40 coach tickets. 3

The company makes a profit of \$225 for each coach ticket and \$200 for each first-class ticket. At most, the plane has a capacity of 150 travelers. How many of

each ticket should be sold in order to maximize profits?

c 3

	Cj	2	3	1	0	0	
CB	Basis	x1	x2	x3	x4	x5	RHS
2	x1	1	0	-1	4	-1	1
3	x2	0	1	2	-1	1	2
	C(bar) row	0	0	-3	-5	-1	Z=8

Maximize Z=Cx ,subject to Ax=b ,x>=0 has the optimal table given above.
Assume (x4,x5) was the initial basis. How much can C2 be changed ,so that current optimal solution remains optimal ?

3 ***Answer any One***

a Solve the following problem by Big -M Method 5

Min z=−3x1+x2+x3

subject to,

x1−2x2+x3<=11

−4x1+x2+2x3>=3

2x1−x3 = −1

where x1,x2,x3>=0

b Using Simplex Method solve the following LPP 5

$Maximize$ z= 3x1+2x2

subject to,

−x1+2x2<=4

3x1+2x2<=14

x1-x2<=3

where x1,x2>=0

And show that alternate optimal solution exist.

4 ***Answer any One***

a Using Revised Simplex method, Solve the following LPP 5

$Maximize\ Z\ =\ 2x_1 - x_2 + x_3$

$subject\ to,$

$$3x_1 + x_2 + x_3 \le 60$$
$$x_1 - x_2 + 2x_3 \le 10$$
$$x_1 + x_2 - x_3 \le 20$$
$$and\ \ x_1, x_2, x_3 \ge 0$$

b Using Dual Simplex method solve the following LPP 5

$Minimize\ Z\ =\ 3x_1 + 2x_2$

$subject\ to,$

$$3x_1 + x_2 \ge 3$$
$$4x_1 + 3x_2 \ge 6$$
$$x_1 + x_2 \le 3$$
$$and\ \ x_1, x_2 \ge 0$$

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Total No. of Pages:-**01**

4th Semester B.Tech. Mid Term Examination 2022-23

Operating Systems(BTCS-T-PC-016)

BRANCH(S) - Computer Engineering,Computer Science and Technology,Computer Science & Engineering

Duration: 01:30 hr Full Marks: 25

1 *Answer All*

a Differentiate between a process and a program. 1

b What do you mean by Spooling in Batch Processing Operating System? 1

c What do you mean by system call? How it is different from a library call? 1

d Enlist the reasons behind process suspension. 1

e A solution to the problem of indefinite blockage of low-priority processes is _______. 1
(i) Priority Scheduling
(ii) Paging
(iii) Aging
(iv) None of these

f Differentiate between Cooperative and Independent processes. 1

2 *Answer any Three*

a What are the five major activities of an operating system with respect to file management? 3

b What are the basic functions of an operating system? 3

c Why thread is called a light weight process? What resources are used when a thread is created? Which components does a thread share with other threads of the same process? 3

d Describe the advantages of co-operating process. 3

3 *Answer any One*

a Explain the different services provided by operating system to the users. 5

b What is the need of an Operating System? Discuss about any three types of Operating Systems with their advantages and disadvantages. 5

4 *Answer any One*

a What are the two basic models of inter process communication (IPC)? Briefly explain each one of them and mention their strengths and weaknesses. 5

b Consider the following set of processes with their CPU burst time, arrival time given in milliseconds and priority. 5

Process	CPU Burst Time	Arrival Time	Priority
P1	3	0	1
P2	2	1	0
P3	4	3	2
P4	5	4	0
P5	3	5	1

Draw three Gantt charts for execution of the processes using SRTF, RR (time quantum = 2), and Preemptive Priority scheduling. Separately compute the average response time, average waiting time of the processes on execution of the three algorithms.

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4th Semester B.Tech Mid-Term Exam 2023-24

Operating Systems(BTCS-T-PC-016)

BRANCH(S) - Computer Engineering,Computer Science and Technology,Computer Science & Engineering

SIC No.-

Duration: 01:30 | Total No. of Pages:-01 | Full Marks: 25

1 ***Answer All***

a Differentiate between multiprogramming and timesharing systems. 1

b Explain the main purpose of an operating system. 1

c What do you mean by Spooling in Batch Processing Operating System ? 1

d What is a Dispatcher? How it works with the scheduler? 1

e Which section is shared by a Process and its thread? 1
(i) Stack
(ii) Register
(iii) Code
(iv) Both (i) and (ii)

f Enlist the reasons behind process suspension. 1

2 ***Answer All***

a What are the five major activities of an operating system with respect to file management ? 3

b Define context switching. With a suitable example explain context switching mechanism among several processes. 3

c Differentiate between independent and co-operative processes with their pros and cons. 3

3 ***Answer any One***

a Explain the structure of the operating systems with suitable diagram. Explain the role of operating system from system and user point of view 5

b What is an operating system? Explain the different services provided by an operating system. 5

4 ***Answer any One***

a What are the two basic models of inter process communication (IPC)? Briefly explain each one of them and mention their strengths and weaknesses. 5

b Draw the Gantt chart and find out the average waiting time and average turnaround time using FCFS, SJF & SRTF scheduling algorithms. 5

Process	Burst Time	Arrival Time
P1	9	1
P2	11	0
P3	3	2
P4	5	4